Q4	Enable and test firewall rules (UFW / iptables)	5 Marks

Objective: Configure a host firewall with a default-deny inbound policy and allow only the services you need.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	CONFIGURE	Set the default policy to deny inbound traffic.	`sudo ufw default deny incoming`	Default deny set
2	CONFIGURE	Allow outbound traffic by default.	`sudo ufw default allow outgoing`	Default allow set
3	CONFIGURE	Allow only required inbound services (e.g. SSH).	`sudo ufw allow 22/tcp`	Rule added
4	EXECUTE	Enable the firewall.	`sudo ufw enable`	Firewall active
5	TEST	Verify the rule set.	`sudo ufw status verbose`	Rules listed

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to lab-vm

After setting a default-deny inbound policy and allowing only SSH, the firewall status shows a small, deliberate rule set instead of everything being open.

Direction	Default policy	Explicit allow	Result
Inbound	**Deny**	22/tcp (SSH)	Only SSH reachable
Outbound	**Allow**	-	Normal traffic works

Sample output:

```
$ sudo ufw status verbose
Status: active
Default: deny (incoming), allow (outgoing)
22/tcp                     ALLOW IN    Anywhere
```

Reusable Template

Firewall rule record template

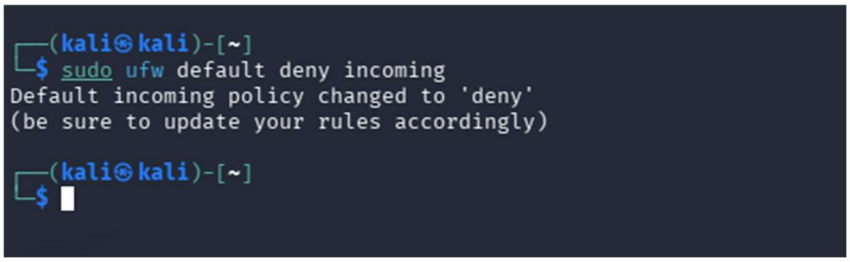
Port / service	Action	Reason it is needed
22/tcp (SSH)	Allow	Needed for secure remote login/administration of the VM
80/tcp(HTTP)	Allow	To serve web pages

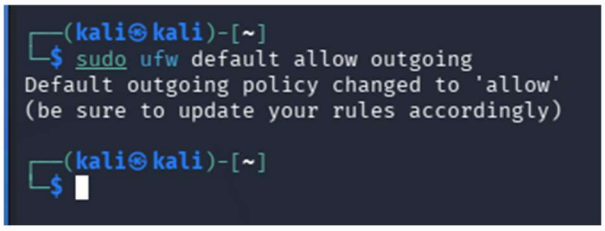
Your Output / Screenshot (paste result here):

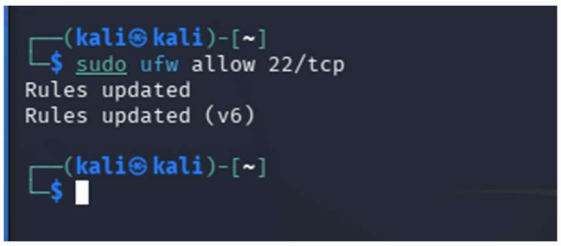
Status: active

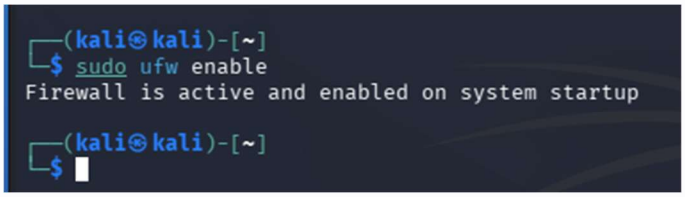
Default: deny (incoming), allow (outgoing)

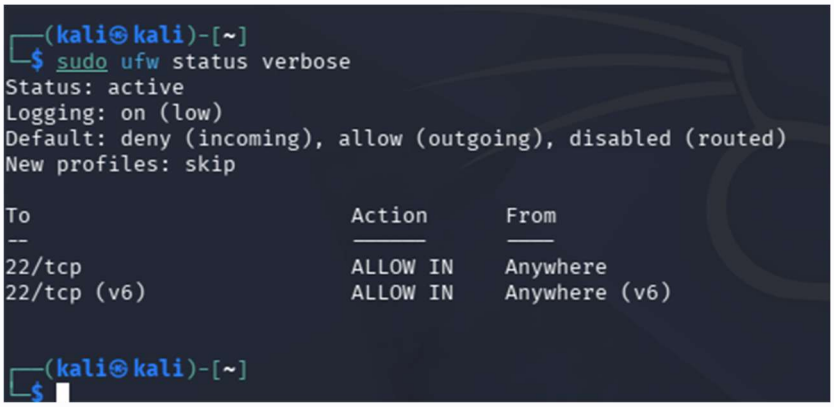
22/tcp ALLOW IN Anywhere











A default-deny inbound policy blocks everything not explicitly permitted, reducing the attack surface. Only SSH (22/tcp) is allowed, while outbound traffic works normally. ufw status verbose confirms the firewall is active with the intended rules.